Statements are tagged with the status of their proof: *Verified* means formalized in Lean 4 against Mathlib with no `sorry` and no axioms beyond `propext`, `Classical.choice` and `Quot.sound`; *Computed* means established by a finite computation, with the generating script cited. An untagged theorem, lemma, proposition or corollary carries a complete proof in the text and no formalization. Conjectures are labelled as such and are proved nowhere here. Definitions carry no status.

A VERIFIED COVERING FRAMEWORK FOR THE LONELY RUNNER PROBLEM IN BOUNDED WINDOWS

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Abstract. We develop, and verify in Lean 4, a covering framework for the lonely runner conjecture restricted to bounded velocity windows. Everything labelled verified below is formalized with no axioms beyond `propext`, `Classical.choice` and `Quot.sound`. First, the window statement is equivalent to the assertion that a hitting number attains a Dirichlet ceiling; this recasting is known in substance and we prove both directions inside the proof assistant. Second, we introduce a surplus normalization of the cover-capacity certificate whose budget is the constant 2, independent of the number of velocities, and prove it equivalent to the usual capacity condition. Third, we classify the bad sets of size at most three into two singleton families, four pair families and three triple families, and prove each of them; the singleton and pair families are exhaustive for every $m \le 260$, the triple families are not, and we exhibit minimal triples outside them. We also prove that the complementary pair family is confined to exactly two moduli. Fourth, we prove a dichotomy on residues: if two velocities are simultaneously close to an integer multiple of the period, then so is their difference or their sum. It yields two of the three shapes a bad pair can take, uniformly in m, and the third follows from an identity between the two residues which we prove by a lattice argument, so that the classification is complete: a bad pair is $\{a, 2a\}$ or has $a + b \in \{2m + 1, 2m + 2\}$ with $\gcd(a, a + b) = 1$. It also shows that the top third of the window carries no bad set, so the bad-pair graph has an independent set of size $m - 1$ for every m.

As an application of the associated decision procedure we decide the window $\{1, \dots, 2m\}$ by exhaustive computation for every $m \le 63$, extending to that range the statement previously verified for $m \le 24$; the extension is computational and is not itself formalized. The same range is then obtained a second time from the nine families alone, with the exhaustive enumeration discarded, so that every set entering the search is one that we have proved to be a bad set. We emphasize that all results are restricted to velocities bounded by $2m$; the unrestricted conjecture is known for at most thirteen runners by different methods, and the two are not comparable. We also record five closed approaches to the uniform statement, including a counterexample at $m = 19$ to the block template suggested by the smaller cases, and counterexamples at $m = 150, 318, 486, 570$ to a regularity of the bad-pair graph that holds at every one of the thirty-three composite moduli with $m \le 63$.

1. Introduction

Let $v_1, \dots, v_m$ be distinct positive integers and write $\|x\|$ for the distance from x to the nearest integer. Put

$$\mathrm{ML}(v_1, \dots, v_m) \;=\; \sup_{t \in \mathbb{R}} \min_{1 \le i \le m} \|v_i t\| \,.$$

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The lonely runner conjecture, due to Wills and independently to Cusick in the language of view obstruction, asserts

$$\mathrm{ML}(v_1, \dots, v_m) \;\geq\; \frac{1}{m+1} \tag{1}$$

for every such tuple. In the runner formulation there are $m+1$ runners, m velocities relative to the observer. The bound is sharp: $v_i = i$ gives equality, and Dirichlet's approximation theorem shows $\mathrm{ML}(1, \dots, m) \leq 1/(m+1)$, so (1) is the statement that a Dirichlet ceiling is attained. See Perarnau and Serra [13] for a survey.

The conjecture is settled for small numbers of runners. At the time of writing it is known for at most thirteen runners, equivalently for $m \leq 12$ velocities in the normalization above, by Rosenfeld [15, 16], Trakulthongchai [19], and Sungkawichai and Trakulthongchai [17]; all of these are computer assisted and descend from a prime divisibility criterion. We state the convention explicitly because the literature is divided between counting runners and counting velocities.

A separate line of work restricts the velocities to a window. Pomerance [14] states that it remains a challenge to prove (1) when $\{v_1, \dots, v_m\} \subseteq \{1, \dots, 2m\}$, and the same restricted problem appears as Problem 6 of [13]. All results in this paper concern that restricted problem, and we state the restriction with every claim. The restricted and unrestricted problems are not comparable, and the method here is not competitive on the unrestricted one; see Remark 1.5.

Substantial parts of this window are already closed, and we state which, since our contribution is what remains. Tao [18] proved that $v_m < 1.2m$ implies (1), uniformly in m [13, Theorem 22]; this closes the lower three fifths of the window outright and is the result that prompts the question of raising 1.2 to 2. Bohman and Peng [4], via coprime mappings, proved that for sufficiently large m the range $m < v_m \leq 2m - \exp(c(\log\log m)^2)$ is also closed, and Pomerance [14] improved the error term to $c(\log m)^2$; both are ineffective in m and so say nothing about any concrete m. Pandey [7] proved (1) for every m-subset of an arithmetic progression $\{a, a+d, \dots, a+(k-1)d\}$ of length $k \leq 2m-3$ with a and d not both 1; taking $d = 1$ this closes every velocity set inside the window whose least element exceeds 1 and whose span $v_m - v_1 + 1$ is at most $2m-3$.

What survives all of these at a given m is the set of $V \subseteq [1, 2m]$ with $v_m \geq 1.2m$ and with either $v_1 = 1$ or $v_m - v_1 + 1 \geq 2m - 2$. That residual contains the extremal configurations: the tight instance $\{1, \dots, m\}$ has $v_1 = 1$, and the tight instances with $v_m = 2m - \Theta(\log m)$ noted in [13, §8.2] nearly span the window. The results above are all of the form "(1) holds for velocity sets satisfying a structural condition". None of them decides the window at a fixed m, that is, settles (1) for *every* m-subset of $[1, 2m]$ simultaneously, and it is that decision which the present method performs.

1.1. **Results.** Throughout, a claim is labelled *verified* if it is formalized in Lean 4 with a clean build, no `sorry`, and no axioms beyond `propext`, `Classical.choice` and `Quot.sound`; *proved* if a complete human proof is given but no formalization; *computed* if it rests on an exhaustive machine enumeration that is not itself formalized; and *conjectural* otherwise. Section 8 gives the correspondence between numbered statements and Lean declarations.

Fix $m \geq 1$ and let $t = m + 1$. For coprime integers p, q with $q > 0$ define the *bad set* of the time p/q inside a window $[1, N]$ by

$$B_N(p, q) \;=\; \big\{\, x \in [1, N] \;:\; \|xp/q\| < 1/t \,\big\},$$

and let $\tau(N)$ be the least cardinality of a subset of $[1, N]$ meeting every $B_N(p, q)$.

Theorem 1.1 (Verified; see Section 3)**.** *For $N \geq m+1$, inequality* (1) *holds for every m-element velocity set contained in $[1, N]$ if and only if $\tau(N) > m$. Moreover $\tau(N) \leq m+1$ for every N, so the window statement is equivalent to $\tau(N) = m+1$ exactly.*

The content of Theorem 1.1 is known. The covering formulation of (1) is standard [18]; that an optimal time may be taken of the form $\ell/(v_i + v_j)$ is elementary and is used throughout the literature without attribution [13, eq. (4)], while the resulting replacement of the supremum by a finite maximum is credited to Haralambis [11, Remark 1], with a proof in [10, Theorem 6]; and Rosenfeld [15, §6, Lemma 7] already runs exactly the resulting finite set-cover search inside a velocity window. The upper bound $\tau(N) \leq m+1$ is the classical Dirichlet observation recorded in [13, §4]. Our contribution to Theorem 1.1 is the formalization and the naming of τ, not the mathematics.

The next statement is a normalization of the cover-capacity certificate of Section 4. It is a change of variables and we claim nothing for it beyond the bookkeeping: the point is that the threshold becomes the constant 2 instead of the m-dependent $m-1$, which is what makes the parts reusable across m.

Theorem 1.2 (Verified; Lemma 4.1)**.** *Let $U_1, \ldots, U_r$ cover $[1, 2m]$, and suppose that for each i every subset of U_i with more than c_i elements contains a bad set. If $\sum_i |U_i| = 2m$ and*

$$\sum_{i=1}^{r} \mathrm{srp}_i \;\geq\; 2, \qquad \mathrm{srp}_i := |U_i| - 2c_i,$$

then (1) *holds for every m-element velocity set contained in $[1, 2m]$.*

The point is the quantifier. The usual condition $\sum_i c_i < m$ has a budget depending on m, so a certificate must be rebuilt for each m; the surplus budget is the constant 2 at every m. The two singleton bad sets supply exactly two units of surplus, and by Proposition 4.2 they are the entire budget available.

Theorem 1.3 (Computed; Families verified, exhaustiveness computed; Section 5)**.** *Nine explicitly parameterized families of bad sets are proved: two of singletons, four of pairs and three of triples. The complementary pair family occurs at exactly two moduli, namely $q = 2m+1$ and $q = 2m+2$, and at no others. The pair families are exhaustive for* every *m: an exact bad pair $a < b$ satisfies $b = 2a$ or $a+b > 2m$, so it is* DBL *or* CMP *(Lean: `badpair_doubling_or_large`, with no hypothesis on the modulus). For singletons the exhaustiveness is computed, over $4 \leq m \leq 260$. The triple families are* not *exhaustive, and Proposition 5.2 exhibits minimal bad triples outside them.*

The families themselves, the two-moduli confinement and the exhaustiveness at size two are verified; the exhaustiveness at size one is computed, over a range wider than any other computation in this paper. The size-two half is unconditional in the modulus, which is why it needs no range at all; the size-one half still rests on a sweep. We separate the two sizes because the distinction is load-bearing: the argument that a particular element closes the independence gap uses only the pair families, whereas the decision procedure of Theorem 6.1 uses all nine and, as explained in Section 5, is unaffected by the triples that the classification misses.

Theorem 1.4 (Computed; Section 6)**.** *For every m with $4 \leq m \leq 63$, inequality* (1) *holds for every m-element velocity set contained in $[1, 2m]$.*

The previously verified range for this restricted problem was $m \leq 24$. Theorem 1.4 is obtained from Theorem 1.1 by deciding $\tau(2m) > m$ directly, which is a decision problem rather than a search for a certificate, and is correspondingly cheaper. The verdicts for $m \leq 24$

reproduce the previously verified range, and the range $25 \le m \le 63$ is new. We stress that Theorem 1.4 is computed and not verified: lifting it to a Lean proof requires a certificate for each m, and producing one is an open problem for each m (Remark 6.3).

The computation admits a sharper description. Because τ is monotone in the family of sets to be hit, the decision may be run against the nine families of Theorem 1.3 alone, with the exhaustive enumeration discarded; it then still exhausts for every $m \le 63$ (Theorem 6.1). On that route the exhaustiveness clause of Theorem 1.3 is not used, every set fed to the search is one of the verified families, and the only unformalized ingredient left in Theorem 1.4 is the finite search itself. That every generated set really is a bad set, which is what soundness needs, is certified constructively for all $m \le 250$; see Section 6.

Finally, Section 7 records five approaches to the uniform statement that are now closed, with the reasons. Among them is a counterexample at $m = 19$ to the block template that the cases $m = 10, 13, 16$ suggest.

1.2. **What is not claimed.** The uniform statement, that (1) holds for every m and every velocity set inside $[1, 2m]$, remains open and nothing here approaches it. The framework certifies individual m; it has no known uniform selection mechanism, and Section 7 explains why the natural candidates fail.

Remark 1.5. The efficiency of the method is governed by the minimum bad-set size, which by Theorem 2.2 equals $\lfloor N/(m+1) \rfloor$. In a window $N = Bm$ with B bounded this is a small constant and the decision procedure is fast; when N/m grows the minimum bad set occupies a constant fraction of the window and the procedure degenerates. Applying the window reduction of [12], which permits taking $N = \binom{m+1}{2}^{m-1}$, the method reproves the unrestricted conjecture for at most four runners and is already impractical at five. It is therefore far behind the prime divisibility sieve on the unrestricted problem, and its regime is the bounded window.

2. Setup

We work with the integer form of the badness predicate, which is decidable and is what the formalization manipulates.

Definition 2.1. For $k, x, p, q \in \mathbb{N}$ with $q > 0$ set

$$\mathrm{good}_k(x,p,q) \;=\; \big[\, q \le (k+2)\min\big(xp \bmod q,\; q - (xp \bmod q)\big)\,\big].$$

We say x is *bad* at (p, q) when $\mathrm{good}_k(x, p, q)$ fails.

Writing $m = k + 1$, badness at (p, q) is exactly $\|xp/q\| < 1/(m+1)$, so $B_N(p,q) = \{x \in [1, N] : x \text{ bad at } (p,q)\}$. A velocity set V with $|V| = m$ fails (1) precisely when every bad set meets V, because the supremum defining ML is attained at a rational time whose denominator is a pairwise sum of velocities [13, eq. (4)], [11, 10]. This is the input to Theorem 1.1.

The following counting bound is used throughout and is sharp.

Theorem 2.2 (Verified)**.** *For all k, N and all p, q with $q > 0$,*

$$\big|\{x \in [1,N] : x \text{ bad at } (p,q)\}\big| \;\ge\; \Big\lfloor \frac{N}{k+2} \Big\rfloor.$$

Proof. Distribute the $N + 1$ points $\{0 \cdot p/q\}, \dots, \{N \cdot p/q\}$ among the $k + 2$ half-open boxes $[j/(k+2), (j+1)/(k+2))$. Some box contains at least $\lfloor N/(k+2) \rfloor + 1$ of them. Subtracting the least index in that box from the others gives $\lfloor N/(k+2) \rfloor$ distinct values in $[1, N]$, each bad, since two points of a common half-open box of length $1/(k+2)$ differ by less than $1/(k+2)$. □

Theorem 2.2 is classical. It is the Dirichlet box principle with B points per box in place of two, and also follows from van der Corput's generalization of Minkowski's theorem applied to $\{|x| < N + \delta,\ |y| < 1/(k+2)\}$ and the unimodular lattice $\{(v, vp/q - u)\}$; see [9]. The same inequality appears inside a proof in [3, eq. (1.34)], there under a linear independence hypothesis that excludes the rational times at which it is sharp. It is sharp: at $p/q = 1/(k+2)$ the bad set is the set of multiples of $k+2$.

3. The hitting formulation

Theorem 3.1 (Verified)**.** *Every bad set meets* $[1, k+2]$*. Consequently* $\tau(N) \le k+2 = m+1$ *for every* $N \ge m+1$*.*

Proof. Apply Theorem 2.2 with $N = k+2$; the floor is 1, so the bad set of any time has an element in $[1, k+2]$. Hence $[1, m+1]$ is itself a hitting set. □

Theorem 3.2 (Verified)**.** *Fix* $N \ge m+1$*.*

(1) *If every subset of* $[1, N]$ *with at most* m *elements misses some bad set, then every* m*-element velocity set inside* $[1, N]$ *satisfies* (1)*.*
(2) *Conversely, if some* $T \subseteq [1, N]$ *with* $|T| \le m$ *meets every bad set, then there are* m *distinct positive velocities inside* $[1, N]$ *violating* (1)*.*

In particular the window statement for $[1, N]$ *is equivalent to* $\tau(N) > m$*, and with Theorem 3.1 to* $\tau(N) = m+1$*.*

Proof. For (1): a velocity set violating (1) meets every bad set, so it is itself a hitting set of size m; the hypothesis applied to it produces a time at which every velocity is good, which contradicts the violation. For (2): extend T to exactly m distinct elements of $[1, N]$, which is possible since $N \ge m$; the extension still meets every bad set, and a family meeting every bad set has no good time, hence violates (1), by the attainment of the supremum defining ML at a rational time with denominator a pairwise sum of velocities, recorded above with references [13, eq. (4)], [11, 10]. □

Remark 3.3. The margin in Theorem 3.2 is always exactly one, by Theorem 3.1. This is the tight instance $\{1, \dots, m\}$ restated in hitting language and should not be read as a structural discovery; the ceiling is classical [13, §4]. It does explain why every certificate in this framework is tight, which is otherwise a recurring surprise.

4. The surplus certificate

For $U \subseteq [1, 2m]$ let $\mathrm{cap}(U)$ denote the largest size of a subset of U containing no bad set, equivalently $|U| - \tau(U)$ where $\tau(U)$ is the least size of a subset of U meeting every bad set contained in U.

The cover-capacity certificate, in the form we use, states that if $U_1, \dots, U_r$ cover $[1, 2m]$ and every subset of U_i with more than c_i elements contains a bad set, then $\sum_i c_i < m$ implies the window statement. The mechanism is that the complement of a violating velocity set is an independent set of size m inside the window, and the parts bound it piecewise. At a general window height Bm the complement has $(B-1)m$ elements and the condition becomes $\sum_i c_i < (B-1)m$; both forms are verified. The soundness of testing a weighted combination of times, equivalently a finitely supported measure, against a family of velocities is Bedert's [1, Lemma 2.1]; we claim no priority for it.

Note that the c_i are upper bounds supplied with the parts, so $c_i \ge \mathrm{cap}(U_i)$, with equality only when the bound is exact; the surplus below is computed from the c_i and not from cap.

Note also that $\sum_i |U_i| = 2m$ together with the covering condition forces the U_i to partition $[1, 2m]$.

Lemma 4.1 (Verified)**.** *Set* $\mathrm{srp}_i := |U_i| - 2c_i$*. Under the hypotheses above with* $\sum_i |U_i| = 2m$*, the condition* $\sum_i c_i < m$ *is equivalent to* $\sum_i \mathrm{srp}_i \geq 2$*.*

Proof. $\sum_i \mathrm{srp}_i = \sum_i |U_i| - 2\sum_i c_i = 2m - 2\sum_i c_i$, and $2m - 2\sum_i c_i \geq 2$ is $\sum_i c_i \leq m - 1$. □

This is arithmetic, not mathematics: it is the same condition written so that the threshold no longer depends on m. We state it separately only because the search consumes it in that form.

The identity is immediate; its value is that the budget no longer depends on m. A certificate becomes: cover the window by parts of nonnegative surplus and locate two units of surplus.

Proposition 4.2 (Proved)**.** *For every* $m \geq 1$ *the only bad sets of size one inside* $[1, 2m]$ *are* $\{1\}$ *and* $\{m + 1\}$*.*

Proof. Fix α and let $\varepsilon = 1/(m + 1)$. Order the $2m + 1$ points $\{x\alpha\}$, $x = 0, \ldots, 2m$, on the circle and let s be the least index difference realised by an adjacent pair, with gap u. The gaps sum to 1; every gap whose index difference is not s has length at least ε, since a shorter one would exhibit a bad singleton of smaller index difference; and at most $2m + 1 - s$ gaps have difference s. Hence $s\varepsilon \leq 1$, so $s \leq m + 1$.

If $s \leq m$ then $2s$ lies in the window, and $2s$ good forces $u \geq \varepsilon/2$. The points are then distinct and every pair $\{x, x + s\}$ is adjacent with gap exactly u, so $1 \geq (2m + 1 - s)\varepsilon/2 + s\varepsilon$, which forces $s = 1$. Thus $s \in \{1, m + 1\}$, and $s = m + 1$ requires $(m + 1)\alpha \in \mathbb{Z}$. Both are realised, at α near 0 and at $\alpha = 1/(m + 1)$ respectively. No approximation theorem is used; the count is the mechanism behind the three-distance theorem. Verified independently by exhaustive breakpoint sweep in exact integer arithmetic for $2 \leq m \leq 400$, with a negative control at threshold $1/(m + 2)$ that fails at every m. □

Proposition 4.3 (Computed)**.** *The number of bad sets of the minimum size* $B - 1$ *inside* $[1, Bm]$ *is exactly two for every* $B \in \{3, 4, 5\}$ *and every* m *in the computed range. (The range of* m *is not recorded and the generating program is not archived; HEURISTIC.) Together with Proposition 4.2, which settles* $B = 2$ *for every* m*, the surplus available from minimum-size parts is exactly two.*

Parts of nonnegative surplus known to us are: a singleton bad set, surplus +1; a bad pair, surplus 0; an odd cycle of bad pairs of length $2\ell + 1$, whose vertex cover number is $\ell + 1$ and independence number ℓ, surplus +1; and a four-element part all four of whose triples are bad, surplus 0.

Proposition 4.4 (Verified)**.** *Let* $S_1, \ldots, S_{m+1}$ *be pairwise disjoint subsets of* $[1, Bm]$*, each of size at least* $B - 1$*. Then at least* B *of them have size exactly* $B - 1$*.*

Proof. Let a be the number of parts of size exactly $B - 1$ and b the number of larger parts, so $a + b = m + 1$ and $(B - 1)a + Bb \leq Bm$. Substituting $a = m + 1 - b$ gives $b \leq m - B + 1$, hence $a \geq B$. □

Proposition 4.4 combined with Theorem 2.2 and Proposition 4.3 shows that a certificate assembled from pairwise disjoint bad sets exists only at $B = 2$: it would require B minimum-size bad sets, and only two exist. At $B = 2$ the requirement is exactly two and exactly two are available, which is why the dense-window certificates are tight and why the two free singletons are structurally necessary rather than convenient.

5. The vocabulary of small bad sets

Fix m and write $k = m - 1$. We classify the bad sets of size at most three inside $[1, 2m]$. A bad set is *minimal* if it contains no bad set properly.

The enumeration is exhaustive over all real times, by the following observation. Membership of a fixed x in the bad set changes only where $\|x\alpha\| = 1/(m+1)$, that is at $\alpha = (j(m+1) \pm 1)/(x(m+1))$, a rational of denominator dividing $x(m+1) \le N(m+1)$. Between two consecutive such breakpoints $a/b < c/d$ in lowest terms the mediant $(a+c)/(b+d)$ lies strictly between and has denominator at most $2N(m+1)$. Hence scanning every reduced p/q with $q \le 2N(m+1)$ meets every breakpoint and every interval between consecutive breakpoints, and the resulting list of bad sets is the list occurring for any real time.

Theorem 5.1 (Computed)**.** *For every m with $4 \le m \le 260$, every minimal bad set of size one or two inside $[1, 2m]$ belongs to one of the families $\{1\}$, $\{m+1\}$, MUL, CMP, DBL below. Over that range this was checked against every bad set of size at most three produced by an enumeration exhaustive over all real times, and no further species of size one or two occurs. The program is* `code/covering/clsweep.rs`*: it checks 2 341 337 minimal bad sets of size at most three over $4 \le m \le 260$ (the sets of size three that lie in none of the families are the triples of Proposition 5.2) and reports no unclassified singleton or pair. An earlier pair-only program,* `code/covering/paircls.rs`*, gives the same verdict for $4 \le m \le 32$; it stores a pair in a 64-bit mask and is void for $m \ge 33$.*

For triples the corresponding statement is *false*, and we record it as such.

Proposition 5.2 (Computed)**.** *The families SUM, SUM′, DBL_3 do not exhaust the minimal bad triples. Roughly one minimal bad triple in eight lies outside them, at every m from 7 to 260. Two smallest examples: $\{5, 8, 13\}$ at $m = 9$, realized at $61/160$, and $\{6, 12, 18\}$ at $m = 10$, realized at $34/209$; both are minimal, and neither satisfies the hypotheses of any of the three propositions. The program* `code/covering/bad_triple_classify.rs` *enumerates all bad sets of size at most three over all real times and compares them as sets with the predicted list, with 0 discrepancies, for $4 \le m \le 260$: it finds 2 272 034 minimal triples, of which 294 170 lie outside the three families (types S3 and CHN of the exact classification), at least one outside at every m from 7 to 260, with a fraction between 0.12 and 0.19 at each such m.*

The two failures are of different kinds and both are instructive. The triple $\{5, 8, 13\}$ has the shape $\{a, b, a+b\}$ with a, b coprime, but sits on the boundary of family SUM, where $2a+b = 18$ is not strictly greater than $2m = 18$, and it is too low for SUM′, which needs $b \ge m+1$. The triple $\{6, 12, 18\}$ is the set of multiples of 6 inside the window, and it has both the SUM shape and the DBL_3 shape, but every family that could carry it requires a modular inverse and $\gcd(6, 12) = 6$. A classification by shape alone would count both as covered; the families are defined by their hypotheses, and it is the hypotheses that fail.

We can prove two of the three shapes for pairs, uniformly in m, and it is worth isolating what the obstruction to the third one is.

Theorem 5.3 (Verified)**.** *Let $a < b$ both be bad at the time p/q. Then $b - a$ is bad or $a + b$ is bad.*

Proof. Write $r_x = xp \bmod q$ and regard r_x as a point on a circle of circumference q; badness of x says r_x lies within $q/(m+1)$ of 0, on one side or the other. If r_a and r_b lie on the same side, the distance from 0 of $r_b - r_a$ is at most the larger of the two distances, so $b - a$ is bad; if on opposite sides, the same holds for $r_a + r_b$, so $a + b$ is bad. □

Corollary 5.4 (Verified)**.** *If the bad set of p/q inside $[1, 2m]$ is exactly $\{a, b\}$ with $1 \le a < b \le 2m$, then $b = 2a$ or $a + b > 2m$.*

Proof. By Theorem 5.3 one of $b - a$, $a + b$ is bad. In the first case $b - a$ lies in $[1, 2m]$ and is smaller than b, so it equals a and $b = 2a$. In the second, $a + b$ exceeds b, so it cannot be one of the two bad values, and therefore cannot lie in the window at all. □

This is the doubling shape and the lower half of the complementary shape. The remaining gap is the upper bound $a+b \le 2m+2$ *in the non-doubling case*, and the restriction is essential rather than cosmetic: $\{m, 2m\}$ is a bad set for every m, by Proposition 5.10 at $d = m$, and it has $a + b = 3m$, which exceeds $2m + 2$ as soon as $m \ge 3$. Doubling pairs are already covered by the first disjunct and must be excluded from the second.

That remaining bound holds as soon as the pair is realized at one particular modulus.

Theorem 5.5 (Verified)**.** *Let $a < b \le 2m$ with $b \ne 2a$, let $ap \equiv 1 \pmod{a + b}$, and suppose the bad set of $p/(a + b)$ inside $[1, 2m]$ is exactly $\{a, b\}$. Then $a + b \le 2m + 2$.*

Proof. Put $q = a + b$. Since $b = q - a$, its residue is -1, and $b - a = q - 2a$ has residue -2. So $b - a$ is bad exactly when $2(m + 1) < q$. But $1 \le b - a \le 2m$, and $b - a$ is neither a, which would force $b = 2a$, nor b, which would force $a = 0$; so $b - a$ bad contradicts the bad set being $\{a, b\}$. Hence $q \le 2m + 2$. □

The hypothesis that the pair is realized at that particular modulus can be weakened to an identity at whatever modulus is given. Fix a time p/q whose bad set inside $[1, 2m]$ is exactly $\{a, b\}$, and suppose the two residues lie on opposite sides of zero: write u for the residue of ap modulo q and $q - v$ for the residue of bp, with $0 < u$ and $0 < v < q$.

Theorem 5.6 (Verified)**.** *With that notation, if*

$$av + bu = q,$$

then $a + b \le 2m + 2$.

Proof. Write $t = m + 1$. Since $b \ne 2a$ and $a \ge 1$, the element $b - a$ lies in $[1, 2m]$ and equals neither a nor b, so it is good. Its residue is $-(u + v)$, whence $q \le (u + v)t$. The identity then gives $a(u + v) < av + bu = q \le (u + v)t$, so $a \le t - 1 = m$ and therefore $2a \le 2m$. As $2a$ is neither a nor b, it is good as well, and its residue is $2u$, whence $q \le 2ut$. Writing $b = a + d$ with $d \ge 1$,

$$q(2a + d) = 2aq + dq \le 2a(u + v)t + d \cdot 2ut = 2t\big(av + (a + d)u\big) = 2tq,$$

and cancelling q gives $a + b = 2a + d \le 2t = 2m + 2$. □

Neither goodness fact is an extra hypothesis; both are forced by $\{a, b\}$ being the whole bad set, and both are derived from that in the formalization. Nothing is required of v beyond positivity. The symmetric partner $q \le 2vt$ of the second fact is false, failing in 3280 of the 11528 instances below (program `code/covering/identchk.rs`, $4 \le m \le 90$, which also fails the second known-false control $a+b \le 2t-1$ in 1601 instances, so that the verdict of 0 failures for the identity is not vacuous), and the argument does not use it.

Theorem 5.6 subsumes Theorem 5.5: at $q = a + b$ with $ap \equiv 1$ the residue of b is -1, so $u = v = 1$ and the identity reads $a + b = q$. The identity is the classical relation between two consecutive best approximations, and it holds.

Theorem 5.7 (Verified)**.** *With the notation above, $av + bu = q$, and $\gcd(a, b) = 1$.*

Proof. Let $\Lambda = \{(x, s) \in \mathbb{Z}^2 : s \equiv xp \pmod q\}$, a subgroup of index q containing (a, u) and $(b, -v)$, and let $\Lambda' = \langle (a, u), (b, -v) \rangle$. The determinant of the second is $av + bu$, so $q \mid av + bu$; write $av + bu = cq$, so that $c \geq 1$ is the index of Λ' in Λ.

Suppose some $(z_0, s_0) \in \Lambda$ lies outside Λ'. Subtracting an element of Λ' puts it in the fundamental parallelogram: $z = \lambda a + \mu b$ and $s = \lambda u - \mu v$ with $\lambda, \mu \in [0, 1)$ not both 0. Then $0 < z < a + b$ and $-v < s < u$, so $|s| < \max(u, v) < q/t$. If $z > 2m$, replace the point by $(a, u) + (b, -v) - (z, s)$, which is again outside Λ', has first coordinate $a + b - z$, and since $a + b \leq 4m - 1$ and $z > 2m$ that is below $2m$. So we may take $1 \leq z \leq 2m$. Then z is bad, hence $z \in \{a, b\}$. If $z = a$ then q divides both $ap - u$ and $ap - s$, hence $u - s$, which lies strictly between 0 and $u + v < q$: impossible. If $z = b$ the same applies to $v + s$. So no such (z_0, s_0) exists, that is $\Lambda' = \Lambda$.

Two points of Λ now give the two conclusions. First, $(1, p) \in \Lambda = \Lambda'$, so 1 is an integer combination of a and b and $\gcd(a, b) = 1$. Second, $(0, q) \in \Lambda$, and $\Lambda' \cap (\{0\} \times \mathbb{Z}) = cq\mathbb{Z}$ because $\gcd(a, b) = 1$; so $cq \mid q$, that is $c = 1$. □

The construction is explicit rather than abstract: for $(0, q)$ the reduced point is $z = (\beta a + \alpha b)/c$ and $s = (\beta u - \alpha v)/c$ with $\beta = b \bmod c$ and $\alpha = -a \bmod c$, and both coordinates are integers by congruence alone, since $\beta a + \alpha b \equiv ba - ab = 0$ and $\beta u - \alpha v \equiv bu + av = cq \equiv 0 \bmod c$. For a general starting point the same formulas apply with c replaced by cq and β, α the residues of $z_0 v + b s_0$ and $u z_0 - a s_0$. That is what is formalized.

Corollary 5.8 (Verified)**.** *Every non-doubling bad pair satisfies $a+b \leq 2m+2$, with no further hypothesis.*

Together with Corollary 5.4, which gives $a + b > 2m$ in the non-doubling case, this pins the complementary shape exactly: $a + b \in \{2m + 1, 2m + 2\}$. The coprimality that the shape alone does not supply is the second half of Theorem 5.7, since $\gcd(a, a + b) = \gcd(a, b)$. The normalization asked about in earlier versions of this problem, that every non-doubling bad pair be realized at the modulus $a + b$, is no longer needed for either.

The identity gives the bound, and the coprimality comes with it. A bad pair may have $a + b = 2m + 1$ with a sharing a factor with the modulus; over $4 \leq m \leq 40$ this happens 25 times, and every one of those pairs is a doubling pair, carried by the first disjunct. (The count 25 was obtained in a session; the generating program is not archived and the count was not regenerated; HEURISTIC.) For non-doubling pairs it does not happen, and that is a consequence rather than an observation: a time realizing the pair at the modulus $a + b$ has $ap \equiv 1$, which exhibits a as a unit. So the normalization yields the families of Section 5 with their hypotheses, not merely the shapes, and that implication is formalized too.

Nothing else in this paper depends on the triple classification being exhaustive. The decision of Theorem 1.4 uses the enumeration, not the families. Theorem 6.1 uses the families, but only in the direction where omitting bad sets is harmless: a smaller family of constraints can only lower the hitting number, so a verdict $\tau > m$ obtained from a subfamily is a verdict for the whole family. The unclassified triples are constraints that were simply not used.

The individual families are verified. We state them with their exact hypotheses; in each case the conclusion is an equality of sets, since a part of a cover must contain a whole bad set.

Proposition 5.9 (Verified: singletons)**.** $\{1\}$ *is the bad set of the time* $1/(2m+2)$*, and* $\{m+1\}$ *is the bad set of the time* $1/(m + 1)$*.*

Proposition 5.10 (Verified: MUL)**.** *Let $2m/3 < d \leq m$. Then the bad set of the time $1/d$ inside $[1, 2m]$ is exactly $\{d, 2d\}$.*

Proof. The band at modulus d has half-width $d/(m+1) < 1$, so x is bad exactly when $d \mid x$; the hypothesis $3d > 2m$ says the third multiple has left the window. □

The number of admissible d is $m - \lfloor 2m/3 \rfloor = \lceil m/3 \rceil$, and this matches the enumerated count of MUL pairs at every m in the computed range.

Proposition 5.11 (Verified: CMP)**.** *Let $q \in \{2m+1,\ 2m+2\}$ and let $ap \equiv 1 \pmod q$ with $2 \le a \le 2m$. Then the bad set of p/q inside $[1, 2m]$ is exactly $\{a,\ q-a\}$. Moreover no other modulus supports a complementary pair: if $\{a, q-a\}$ is a bad set with residues ± 1, then $q > 2m$, since otherwise q itself is bad at residue 0, and $q \le 2m+2$, since otherwise residue ± 2 is bad and the set is larger.*

The case $q = 2m+1$ is the coprimality edge of Bohman and Peng [4, §3], who take the time $s^{-1}/(s+t)$ and observe that no velocity is 0 or ± 1 modulo $s+t$; we cite it and do not claim it. The case $q = 2m+2$ appears not to be recorded. At $a = 1$ the partner $q - 1 = 2m+1$ leaves the window and the set degenerates to the singleton $\{1\}$, consistently with the singleton family.

Proposition 5.12 (Verified: DBL)**.** *Let $2m+2 < q \le 3m+3$, let $ap \equiv -1 \pmod q$, and suppose $2a \le 2m$ and $2m + 2a < q$. Then the bad set of p/q inside $[1, 2m]$ is exactly $\{a, 2a\}$.*

The hypothesis $2m + 2a < q$ pushes the complementary partners $q - a$ and $q - 2a$ out of the window; without it they re-enter and the bad set has four elements.

The classification of pairs has a structural consequence for the graph they form. Write G_m for the graph on $[2, 2m]$, on $2m-1$ vertices, in which a and b are adjacent when $\{a, b\}$ is a bad set. The vertex 1 is omitted because it is a bad singleton. Section 7 works instead with G, the induced subgraph of G_m on $[2, 2m] \setminus \{m+1\}$, on $2m-2$ vertices, which drops the second bad singleton as well. The two are distinct and both are used: triangle freeness is stated for the larger G_m, and the matching and independence results for G, where the candidate complements live.

Theorem 5.13 (Verified)**.** *G_m contains no triangle, for every m.*

Proof. Every edge is a doubling pair $b = 2a$ or a complementary pair with $a+b \in \{2m+1, 2m+2\}$, by Corollary 5.4 and Theorem 5.5. Let $2 \le x < y < z$ be a triangle. If all three edges are complementary, the three sums lie in $\{2m+1, 2m+2\}$, which forces $x, y, z \in [m-1, m+2]$; the only triple there with $x + y \ge 2m+1$ is $\{m, m+1, m+2\}$, and it fails because $y + z = 2m+3$. If exactly one edge doubles, subtracting the two complementary sums gives $x \le 1$. If two edges double, the triangle is $\{x, 2x, 4x\}$ with $5x \in \{2m+1, 2m+2\}$, and the gcd condition then gives $x = 1$. Three doubling edges are inconsistent with $x < y < z$. Every branch contradicts $x \ge 2$. □

Two consequences are used in Section 7. A part of three elements and capacity one requires all three of its pairs to be bad, so no such part exists. And no five-cycle of G_m has all five edges complementary, since two complementary edges at a vertex force its neighbours to differ by at most one, which places all five vertices in a window of width two.

For triples we record the three families and omit the proofs, which are of the same character.

Proposition 5.14 (Verified: SUM, SUM′, SUM″, DBL_3)**.** *Let $a < b$.*

1. *If $m+1 < a+b \le 2m$, $2a+b > 2m$ and $ap \equiv 1 \pmod{a+b}$, the bad set of $p/(a+b)$ is exactly $\{a, b, a+b\}$.*
2. *If $b \ge m+1$, $a+b \le 2m$, $a+2b \le 3m+3$ and $bp \equiv 1 \pmod{a+2b}$, the bad set of $p/(a+2b)$ is exactly $\{a, b, a+b\}$.*

(3) *If* $2a+b \le 2m$, $b \le m$, $a+2b > 2m$, $m+1 < a+b$, $3m+3 < 2a+3b \le 4m+4$ *and* $(a+b)p \equiv 1 \pmod{2a+3b}$, *the bad set of* $p/(2a+3b)$ *is exactly* $\{a, b, a+b\}$. *The residues of* a, b, $a+b$ *at this modulus are* 3, -2, 1.
(4) *If* $2a \le 2m$, $t \le 2m$, $a+t > 2m$, $2m+2 < 2a+t \le 3m+3$ *and* $ap \equiv 1 \pmod{2a+t}$, *the bad set of* $p/(2a+t)$ *is exactly* $\{a, 2a, t\}$.

The third item is family SUM″. It is the branch the enumeration of Section 6 has always used and this proposition did not state; the lower bound $3m+3 < 2a+3b$ is necessary and was absent from the enumeration's predicate. The smallest member carries residue 3, so it is bad only when $3(m+1) < 2a+3b$, while the remaining conditions give only $2a+3b > 3m+1$. Without it the branch admits sets whose smallest member is not bad, of which there are 38 below $m = 60$, the first being $\{3, 4, 7\}$ at $m = 5$, where $2a+3b = 18$ equals $3(m+1)$ and the bad set is $\{4, 7\}$. With it the branch is realizable throughout the tested range, and the enumeration's membership check is unaffected.

Remark 5.15 (Formalization)**.** The side condition $a+t > 2m$ of item (3) need not be stipulated. Write θ_x for the signed distance from $x\alpha$ to the nearest integer, so that x is bad exactly when $|\theta_x| < \varepsilon$, and note that on the window the residues add. For an exact bad set $\{a, 2a, t\}$ the sum $a+t$ exceeds t, so if it were bad and at most $2m$ it would be a member larger than the largest, which is impossible; hence either $a+t > 2m$, or $a+t$ is not bad and the difference $t-a$ is, and $t-a$ lies strictly below t, so it is a or $2a$ and $t = 3a$. Thus $a+t \le 2m$ forces the chain $\{a, 2a, 3a\}$, and item (3), where $t \ne 3a$, must satisfy $a+t > 2m$.

The same argument gives the shape dichotomy: an exact bad set $\{a, b, c\}$ with $a+b \le 2m$ satisfies $c = a+b$ or $b = 2a$. For $a+b > 2m$ the sum branch is vacuous, since badness outside $[1, 2m]$ identifies no member, and the reflection of Theorem 5.7 is required: a reduced lattice point $z = \lambda a + \mu c$ above the window reflects to $a+c-z \in [1, 2m]$ with residue $(1-\lambda)\theta_a + (1-\mu)\theta_c$, and since $z > 2m$ forces $\lambda + \mu > 1$, that residue is bounded by $(2-\lambda-\mu)\max(|\theta_a|, |\theta_c|)$ and is therefore strictly smaller. The reflected point is bad and in range, hence a member.

The determinant argument behind Lemma 3.1 is formalized in the same way. With $\theta_a = a\alpha - j_a$ and $\theta_b = b\alpha - j_b$ the quantity $a\theta_b - b\theta_a$ equals $bj_a - aj_b$, an integer, the α terms cancelling; and it is bounded by $(a+b)\varepsilon$, which stays below 2 on the window. An integer of absolute value below 2 is -1, 0 or 1, and the vanishing case makes the two lifts parallel, so that $\theta_x = x\delta$ with $\delta = \alpha - k$ and every $x \le a$ is bad, forcing the least bad speed to be 1. Hence the determinant is ± 1, which is itself the Bézout relation: $bj_a - aj_b = 1$ reads $a(-j_b) + bj_a = 1$, so $\gcd(a, b) = 1$ and the coefficients are exhibited at once.

These statements are formalized in Lean, without `sorry` and on the standard axioms, in `ShapeTheorem.lean`, `ReflectionResidue.lean` and `LatticeReduction.lean`. The membership of every bad set of size at most three in the five families is checked for $2 \le m \le 150$, over $q \le 2m+2$, by kernel evaluation and for $2 \le m \le 300$, comprising 5,882,334 sets, by an independent exhaustive sweep. The shape dichotomy is proved for $a+b \le 2m$. The complementary case is equivalent to a sharper statement: an exact bad set whose two smallest members form a DBL pair with $3a > 2m$ has no third member. Over $2 \le m \le 150$ there are 194,811 such pairs and none extends (program `code/covering/bt_dbl_extend.rs`), so the statement is non-vacuous, but it is not proved. It cannot be proved from the residue inequalities alone: with $a > 2m/3$ every speed is r, $a+r$ or $2a+r$ with $0 \le r < a$, a third member exists exactly when some r satisfies $|2\theta_a + \theta_r| < \varepsilon$ while r and $a+r$ are not bad, and that system is satisfiable, for instance at $\theta_a = \varepsilon/4$, $\theta_r = -5\varepsilon/4$. The obstruction must come from which residues occur among the available r, which is a three-distance question.

Families (1) and (2) realize the same triple $\{a, b, a+b\}$ at two different moduli; (1) applies when the second periods stay outside the window and (2) when they do not. The distinction is not cosmetic: at $m = 7$ the triple $\{2, 9, 11\}$ fails the hypothesis of (1), since $2a + b = 13 \leq 14$, and the bad set at modulus 11 is in fact $\{2, 9, 11, 13\}$; it is realized by (2) at modulus 20.

Remark 5.16. An empirical observation, computed and not proved, guided the choice of families. Extracting an irredundant subfamily of bad sets that still forces $\tau(2m) > m$, by repeatedly discarding a bad set and re-deciding, produces for every $m \leq 33$ a core containing no bad set of size four or more. When $2m + 1$ is prime the core has exactly $m + 1$ members, namely the two singletons and $m - 1$ pairs forming a perfect matching, which recovers the structure behind the uniform proof for prime $2m + 1$. When $2m + 1$ is composite the core is larger and always contains triples. This is why the classification stops at size three.

6. Decision procedure and computations

Theorem 3.2 turns the window statement into the decision problem: does a hitting set of size m exist? We decide it by branch and bound, seeding the incumbent at $m+1$, branching on the smallest not-yet-hit bad set, and pruning with the bound that pairwise disjoint not-yet-hit bad sets each require a distinct element. Bad sets are enumerated exhaustively as in Section 5 and reduced to minimal ones.

A verdict of the form "no hitting set of size m exists" is robust to under-enumeration: τ is monotone in the family of sets to be hit, so establishing $\tau > m$ for a subfamily establishes it for the whole family. Only a verdict in the other direction would require completeness of the enumeration, and none occurred.

Proof of Theorem 1.4. The procedure was run for $4 \leq m \leq 63$, the ceiling being the width of the machine word used for the window bitmask. In every case the search exhausted without finding a hitting set of size m, whence $\tau(2m) = m + 1$ by Theorem 3.1 and the window statement follows by Theorem 3.2. Total running time was under thirteen minutes, dominated by the enumeration of bad sets rather than by the search. □

The same range from the verified families alone. The monotonicity just used has a consequence worth separating out. Since a verdict $\tau > m$ for a subfamily already gives the window statement, the search does not need the exhaustive enumeration at all: it may be run against the nine families of Section 5 and nothing else. Those families are verified, so on this route the enumeration, which is the one ingredient of Theorem 1.4 that is neither formalized nor independently auditable, is removed entirely.

Theorem 6.1 (Computed, from verified inputs)**.** *For every m with $4 \leq m \leq 63$, the hitting-set search run against the two singleton, four pair and three triple families of Section 5, and against no other set, exhausts without finding a hitting set of size m. Consequently the window statement holds for every such m.*

The soundness of this route depends on an inclusion in one direction only, and it is the opposite of the one in the previous paragraph. Under-generation is harmless. Over-generation is not: a set that is not a bad set is a constraint with no content behind it, and since adding constraints can only raise τ, a single such set could manufacture a false verdict $\tau > m$. What is required is therefore exactly that every generated set is a bad set, and this is certified constructively rather than assumed. Each family member is emitted together with the time p/q named in its proposition, and $B_{2m}(p, q)$ is recomputed from the definition and compared with the generated set. Over $4 \leq m \leq 250$ this was carried out for all 2,772,478 generated members; in every case the two sets were equal, not merely nested. The certificate needs no

enumeration and is therefore available at every m, including those far beyond the range the search itself reaches.

Four controls were run. First, the verdicts for $m \le 24$ agree with the independently verified range. Second, replacing the threshold $1/(m+1)$ by $1/m$ makes the statement false, since $\mathrm{ML}(1,\dots,m) = 1/(m+1)$; the procedure then finds a hitting set, and the witness it returns is $\{1,\dots,m\}$, the classical tight instance. Third, an independent implementation computing $\tau(2m)$ exactly, with a different bound and code path, returns $\tau = m+1$ at $m = 25$ and $m = 26$, in agreement. Fourth, a search in the complementary formulation, looking for an m-set containing no bad set rather than for a hitting set, returns the same verdict at every $m \le 45$; it shares no branching logic with the first implementation.

Remark 6.2. The enumeration of bad sets can be made much cheaper than the scan over all reduced p/q with $q \le 2N(m+1)$, which costs $O(m^4)$ times the window width. Membership of x changes only where $\|x\alpha\| = 1/(m+1)$, that is at $\alpha = (j(m+1)\pm 1)/(x(m+1))$, so there are only $O(m^2)$ breakpoints in $[0,1)$; evaluating the bad set at every breakpoint and at one interior point of every interval between consecutive breakpoints therefore produces the complete family in $O(m^3)$. The breakpoints must be evaluated and not only the intervals: at a breakpoint some x has $\|x\alpha\| = 1/(m+1)$ exactly and is good there while bad on both sides, so the bad set at a breakpoint is a proper subset of both its neighbours, and a smaller bad set is a stronger constraint. Carried out in exact integer arithmetic on rationals, with interior points taken as mediants, this reproduces the family of minimal bad sets produced by the scan exactly, set for set, for every $m \le 26$. With the enumeration no longer the bottleneck, the branch and bound is, and its cost tracks the deficiency rather than m, which is what Proposition 7.2 would lead one to expect: searching against the nine families, $m = 56$ takes under a second and $m = 61$, whose deficiency is 18, takes half an hour.

Remark 6.3. Theorem 1.4 is computed, not verified. The decision procedure establishes the nonexistence of a small hitting set, which is not by itself a certificate that the Lean statement can consume; the verified entry points require a cover with capacities. By Lemma 4.1 and Proposition 4.2, producing such a cover at a given m amounts to partitioning $[1,2m]\setminus\{1,m+1\}$ into parts of surplus zero, and we do not know how to do this uniformly; see Section 7.

A case above the window: $m = 13$ at height three. The decision procedure is not tied to the window $[1,2m]$. Raising the height to $3m$ at $m = 13$ gives a case the literature does not cover, since the conjecture is known for at most 12 velocities and that is independent of height. We record the convention once more because a single velocity separates new from known: m counts velocities and the number of runners is $m+1$, so $m = 13$ is 14 runners.

Theorem 6.4 (Computed)**.** *Every* 13*-element set of velocities inside* $[1,39]$ *satisfies* (1)*.*

The criterion is the hitting formulation of Section 3 applied verbatim at height three. A velocity set is lonely exactly when some time's bad set misses it entirely, so a counterexample is a set meeting every bad set. Over $[1,39]$ at threshold $1/14$ the family of distinct bad sets has 717 members, obtained from the breakpoints together with the reduced mediants between consecutive ones. All $\binom{39}{13} = 8\,122\,425\,444$ velocity sets were tested against it and none meets every member.

The direction of any error in the enumeration is the one that matters. Each member of the family is computed from an explicit rational by the definition, so the family can only omit bad sets, never contain a set that is not bad. An omission would report a lonely set as a counterexample, not the reverse, so a count of zero is safe while a positive count would have required checking each witness by hand.

Two independent checks support the computation. First, the same procedure run at $m = 10, 11, 12$ and the same height $3m$ returns no counterexample in each case, agreeing with the literature, which settles those by other methods. Second, the procedure detects counterexamples when they exist: at threshold $1/13$ rather than $1/14$ the set $\{1, \dots, 13\}$, whose maximum loneliness is exactly $1/14$, is reported, and at threshold $1/10$ in $[1, 20]$ four sets are reported. An earlier version of the program discarded empty bad sets, which are the times at which nothing in the window is bad, and consequently reported $\{1, \dots, 13\}$ as a counterexample at threshold $1/14$; the second check is what caught it.

Theorem 6.4 is computed and not verified, and the obstruction is sharper than the one in Remark 6.3. At height $b + 1$ the budget of Lemma 4.1 becomes $b(k + 1)$, so at $m = 13$ and height three a certificate needs parts covering $[1, 39]$ with capacities summing to at most 25, an average ratio of 0.641. At this height there is no bad singleton and there are only two bad pairs, so the parts available at or below that ratio are scarce: one of size four, 54 of size five, none of size six with capacity three, and eight of size seven, the last all containing $\{1, 2, 14, 28\}$. Every element of $[1, 39]$ lies in some such part, so nothing is excluded outright, yet an exhaustive search over parts of size two through seven finds no cover within budget. The scarce elements, each lying in exactly three candidate parts, are what the search fails on.

7. Closed approaches

We record five approaches to the uniform statement that are now closed, with reasons. Each was pursued to a decision, and the negative results are the durable part of this work.

Bad pairs alone, integrally or fractionally. A cover by the two singletons and $m - 1$ disjoint bad pairs certifies the window, and such a matching exists precisely when $2m + 1$ is prime or equal to 9, which was checked by blossom matching for every $m \le 24$. The fractional relaxation does no better: by half-integrality of the fractional matching polytope a vertex set decomposes into edges and odd cycles exactly when the graph has a fractional perfect matching, and computation over $5 \le m \le 24$ shows the bad-pair graph has one if and only if $2m + 1$ is prime.

Let G be the bad-pair graph on $[1, 2m] \setminus \{1, m + 1\}$, on $n = 2m - 2$ vertices, that is the induced subgraph of G_m obtained by deleting $m + 1$, with Gallai–Edmonds decomposition (D, A, C) and deficiency $\mathrm{def} = n - 2\nu(G)$.

Proposition 7.1 (Proved; Classical)**.** *$\alpha(G) \le m - 1 + \frac{1}{2}\mathrm{def}$, with equality if and only if G is König–Egerváry, that is $\alpha(G) + \nu(G) = n$.*

We claim nothing here: the statement is Gallai's identity combined with the definition of the König–Egerváry property, and we record it only to fix the direction of the inequality, which is the direction that makes it useless on its own.

Proof. Gallai's identity $\alpha(G) + \rho(G) = n$, with ρ the minimum vertex cover, together with $\rho(G) \ge \nu(G)$, gives $\alpha(G) \le n - \nu(G)$, and equality holds exactly when $\rho(G) = \nu(G)$, which is the König–Egerváry property. Substituting $\nu(G) = (n - \mathrm{def})/2$ and $n = 2m - 2$ turns $n - \nu(G)$ into $m - 1 + \frac{1}{2}\mathrm{def}$. □

The inequality runs in the direction that is useless on its own: it caps $\alpha(G)$ rather than forcing it up. What the computation supplies is the equality case.

Proposition 7.2 (Computed)**.** *For every composite $2m + 1$ with $m \le 63$ the bad-pair graph is König–Egerváry, so $\alpha(G) = m - 1 + \frac{1}{2}\mathrm{def}$ exactly. The independence number was computed by exact maximum independent set, by a code path that does not use the deficiency, and the*

deficiency by blossom matching; the identity $\alpha + \nu = n$ was then read off. Consequently $\alpha(G) \geq m$ at each of the thirty-two of those values with $\mathrm{def} \geq 2$*, the one exception being $m = 4$, where $2m + 1 = 9$ and* $\mathrm{def} = 0$*.*

Suppose $\alpha(G) \geq m$ and let S be an independent set of G with $|S| = m$. Its complement $H = [1, 2m] \setminus S$ also has m elements, and it meets every bad set of size at most two: it contains 1 and $m + 1$, which lie outside the ground set of G and so outside S, and it meets every bad pair, since a bad pair inside S would be an edge of G inside an independent set. Hence the hitting number of the subfamily consisting of the bad singletons and the bad pairs is at most m, and no certificate assembled from that subfamily can reach $m + 1$. So at every composite $2m + 1$ with $m \leq 63$ other than $m = 4$, no certificate built from bad pairs alone can work, integrally or fractionally, and the bad triples are doing work that nothing smaller can do.

This is a computed statement, and it cannot be promoted, because the König–Egerváry property is *false* in general.

Proposition 7.3 (Computed)**.** *The bad-pair graph is not König–Egerváry for every m. Over $4 \leq m \leq 900$ it fails at exactly four values, $m = 150, 318, 486, 570$. The least is $m = 150$, where $2m + 1 = 301 = 7 \cdot 43$: there $|V| = 298$, $|E| = 323$ and $\nu = 139$, and no vertex cover of size ν exists.*

All four have $7 \mid 2m + 1$. We state that as an observation and not as a pattern: it is four points, the divisibility is necessary over the tested range but plainly not sufficient, since 128 of the 897 values tested satisfy it and only four fail, and the tightest residue class containing all four still contains six values that do not fail. Whether the failures are finite, sparse, or merely rare is open, and nothing in this paper depends on the answer.

Whether a graph is König–Egerváry is decidable in polynomial time, without computing the independence number, which is what makes a search out to $m = 150$ possible. Fix a maximum matching M. A vertex cover of size $|M|$ must contain exactly one endpoint of each edge of M and no M-exposed vertex, since the $|M|$ disjoint edges already exhaust the budget; one boolean per edge of M then turns the covering requirement into a 2-SAT instance, satisfiable exactly when the graph is König–Egerváry. In the satisfiable case the assignment is an explicit minimum vertex cover, so a positive answer certifies itself. König–Egerváry graphs carry a literature of their own [5, 8, 6]; Proposition 7.3 says the bad-pair graph does not belong to it, so that literature is not a route to a proof here.

The equality $\alpha(G) = m - 1 + \frac{1}{2}\mathrm{def}$ is therefore a feature of small m rather than a theorem, and the negative result above rests on the measured values of $\alpha(G)$ and on nothing else. One half of it is unconditional and worth isolating, since it needs neither computation nor the classification.

Proposition 7.4 (Verified)**.** $\alpha(G) \geq m - 1$ *for every m.*

Proof. By Theorem 3.1 every bad set meets $[1, m + 1]$. Hence no bad set is contained in $[m + 2, 2m]$, so that interval contains no bad set at all; in particular it spans no edge of G, and it is an independent set of G of size $m - 1$. □

The three steps are formalized: that the bad set always escapes the top interval, that no bad set is contained in it, and its specialization to the two-element sets which are the edges of G, together with the cardinality $m - 1$.

The remaining unit, from $m - 1$ to m, reduces to a statement about a single element. Write $T = [m + 2, 2m]$ and let $x \notin T$. Since no bad set is contained in T, every bad set contained in $T \cup \{x\}$ must contain x; this too is verified. So $T \cup \{x\}$, which has m elements, is independent

in G as soon as $\{x\}$ is not a bad set and no $\{x, y\}$ with $y \in T$ is a bad set. Nothing about how bad sets distribute over T enters.

Proposition 7.5 (Verified, given the classification)**.** *Let $2m+1$ be composite and let p be its least prime factor. Assume the classification of Theorem 5.1 in the form that a doubling bad pair $\{a, 2a\}$ satisfies the hypothesis of MUL or of DBL. Then for every $m \geq 13$ the element $x = 2p$ forms no bad pair with any element of T, so $\alpha(G) \geq m$. The exceptions below that range are $m = 7$ and $m = 12$, where some element other than $2p$ is needed, and $m = 4$, where $2m + 1 = 9$ and no element works at all.*

Proof. Since $p^2 \leq 2m + 1$ we have $2p < m + 2$, so $2p$ lies below T and the pair is $\{2p, y\}$ with $y \in T$. If it is complementary at $2m+1$ then a is a unit there, but p divides both $2p$ and $2m+1$; if complementary at $2m + 2$ then 2 divides both. If it is a doubling pair then $y = 4p$. Under MUL, $3 \cdot 2p > 2m$ gives $3p > m$, so $9p^2 > m^2$; with $p^2 \leq 2m + 1$ this forces $m^2 < 18m + 9$, hence $m \leq 18$. Combined with $m \geq 13$ and $3p > m$ this gives $p \geq 5$, while $p^2 \leq 2m + 1 \leq 37$ gives $p \leq 6$, so $p = 5$; then $3p > m$ forces $m \leq 14$ and $5 \mid 2m + 1$ fails, since $2m + 1$ is 27 or 29. Directly: for each m with $13 \leq m \leq 18$ and $2m + 1$ composite, namely $m \in \{13, 16, 17\}$ with least prime factor $3, 3, 5$, the condition $3p > m$ already fails. Under DBL the modulus condition $2m + 2 \cdot 2p < 3m + 3$ gives $4p < m + 3$, while $y = 4p \in T$ needs $4p \geq m + 2$; so $4p = m + 2$, whence $2m + 1 = 8p - 3$ and $p \mid 3$, giving $p = 3$ and $m = 10$. Both are below 13. □

Only $m = 17$ and $m = 24$ are threatened at all in $13 \leq m \leq 28$, and at both the MUL and DBL hypotheses fail. The weaker form of the classification, shape together with coprimality, suffices for $m \geq 29$, where the doubling case is excluded by size alone.

The three failures are of two different kinds, and the distinction matters. At $m = 4$ the deficiency is 0 and $\alpha(G) = m - 1$ exactly, so no element can work and nothing is missing. For prime $2m + 1$ no element works either, again correctly, and for the same reason. At $m = 7$ and $m = 12$, by contrast, $\alpha(G) \geq m$ does hold while no set of the shape $T \cup \{x\}$ attains it, so there the construction is genuinely insufficient rather than merely unlucky.

Proposition 7.6 (Verified)**.** *For $m = 7$ and for $m = 12$, every $x \in [2, m]$ admits a time whose bad set inside the window is a two-element subset of $T \cup \{x\}$. Hence no set of that shape is independent in G at those two values.*

Each of the seventeen blocking pairs is certified by an explicit time, and membership in a bad set is decidable, so the proposition is settled case by case inside the proof assistant rather than read off a table. We formalize it because the shape that fails is as much a part of the record as the shape that works.

Why $2p$ should work is visible from the families of Section 5: $2p$ is a non-unit modulo $2m+1$, so it carries no CMP$(2m + 1)$ pair; it is even while every unit modulo $2m + 2$ is odd, so it carries no CMP$(2m + 2)$ pair; and its doubling partners p and $4p$ are both below $m + 2$ once $p \leq \sqrt{2m + 1}$, so the MUL and DBL pairs through it do not reach T. That argument is not a proof, because it presumes the classification exhaustive, which is Problem 2 of Section 10.

Enumerating the independent sets. Proposition 7.2 makes the residual task explicit: every independent m-set of G must contain a bad set of size at least three. Writing $I(m)$ for the number of independent m-sets, the computed values (program `code/covering/indcount.rs`, $4 \leq m \leq 24$; it returns $I(m) = 0$ at every prime $2m + 1$ in the range) show $I(m)$ is governed by the deficiency rather than by m: $I = 1, 9, 3, 41, 1203$ at def $= 2$ (for $m = 7, 10, 12, 13, 24$), $I = 310, 728$ at def $= 4$, $I = 4093$ at def $= 6$, and $I = 40493$ at def $= 8$. Note $m = 22$, with def $= 8$, has $I = 40493$ while the larger $m = 24$, with def $= 2$, has $I = 1203$.

Over the computed values I grows by roughly an order of magnitude per two units of deficiency, so that $I \approx 10^{\mathrm{def}/2}$ describes the data. If the deficiency is $\Theta(m)$ for infinitely many m, which we expect but do not prove, and if that empirical growth persists, then $I(m)$ is exponential in m along those m, and any argument treating the independent m-sets individually, by enumeration or by case analysis, is not uniform in m. We state this as the heuristic it is; it is a guide to the shape of an admissible proof rather than a proved obstruction, and each of the approaches below violates it.

Disjoint bad sets at greater height. By Proposition 4.4 a certificate of pairwise disjoint bad sets inside $[1, Bm]$ needs B bad sets of the minimum size $B-1$, while by Proposition 4.3 only two exist. Hence this shape of certificate is confined to $B = 2$.

Prime square moduli. The Gallai–Edmonds decomposition of the bad-pair graph satisfies $|D| - |A| = \mathrm{def}$ at all thirty-three composite $2m+1$ with $m \le 63$, so over that range D is an independent set and the obstruction is a Hall violator rather than an odd component. We state the range because it is the whole evidence: the König–Egerváry regularity of Proposition 7.2 was measured over exactly the same values and fails at $m = 150$. One hopes for an arithmetic description of D. Two natural candidates fail: D is not the set of non-units modulo $2m+1$, and the deficiency is not the number of odd doubling chains among the non-units, agreeing in only seven of fourteen computed cases and failing in both directions. For $q = 2m+1 = p^2$ the deficiency takes the values $0, 2, 2, 4$ at $p = 3, 5, 7, 11$, so the family is not uniform either.

The six-three template. Every non-overlapping certificate we verified consists of the two singletons together with $(m-1)/3$ parts of six elements and capacity three, which requires $3 \mid m-1$. This holds at $m = 10, 13, 16$. It fails at $m = 19$: enumerating all six-element subsets of $[1, 38] \setminus \{1, 20\}$ and retaining the 4440 with capacity exactly three, an exhaustive exact-cover search terminates without a partition after 969,154,513 nodes (program `code/covering/tmpl63.rs`; its default node cap is $2 \cdot 10^8$, so the cap must be raised; the same program finds the template at $m = 10, 13, 16$). So the template is a coincidence of small m, not a family.

8. Formalization

The development is in Lean 4 with Mathlib. Every statement labelled verified builds cleanly with no `sorry` and depends only on `propext`, `Classical.choice` and `Quot.sound`; the few statements that use no choice are noted in the source. We are not aware of a previous formalization of any part of the lonely runner conjecture in any proof assistant.

Statement	Lean declaration
Theorem 2.2	`minBadCount`
Theorem 3.1	`bad_meets_initial`
Theorem 3.2(1)	`floor_of_no_hitting_set`
Theorem 3.2(2)	`not_floor_of_hitting_set`
converse witness	`exists_good_time_of_floor`
Lemma 4.1	`floor_of_surplus`
capacity certificate, height B	`floor_of_cover_capacity_height`
Proposition 4.4	`packing_needs_minimal`
Proposition 7.4	`no_badpair_in_top`, `card_top_interval`
supporting steps	`bad_escapes_top`, `no_badset_in_top`
reduction to a single element	`badset_in_top_insert`, `card_top_insert`
Proposition 7.6	`top_insert_blocked_m7`, `top_insert_blocked_m12`
Theorem 5.3	`bad_diff_or_sum`
Corollary 5.4	`badpair_doubling_or_large`
Theorem 5.5	`pairUpper_at_modulus`
Proposition 7.5	`twoP_closes_gap_full`, `twoP_closes_gap`
reduction chain	`pairUpperBound_of_realized`, `pairClassification_of_realized`
with coprimality	`pairClassificationStrong_of_realized`
Proposition 5.10	`badset_mul_pair`
Proposition 5.11, $q = 2m + 1$	`badset_unit_pair`
Proposition 5.11, $q = 2m + 2$	`badset_compl_pair`
Proposition 5.12	`badset_dbl_pair`
Proposition 5.14(1),(2),(3)	`badset_sum_triple`, `badset_sum_triple_high`, `badset_dbl_triple`
all windows imply the conjecture	`lonelyRunner_of_all_windows`
dense window, $2m + 1$ prime	`dense_floor_of_prime`
dense window, $2m + 1$ composite, $m \leq 24$	`dense_floor_m4`, `dense_floor_m7`, …, `dense_floor_m24`
Theorem 5.7, the reflection step (gap G1)	`dbl_reflection`

The programs for the computed statements are in `code/covering` and `code/dense` of the accompanying archive; its `README.md` lists, statement by statement, which counts are regenerated by a program in the archive and which are not.

Statements that are computed or conjectural, rather than verified, are named in Lean as predicates and are never asserted. Naming them fixes their quantifiers and the direction of their inequalities before any effort is spent on a proof, and a statement that cannot be typed out correctly is not yet understood; but a proof obligation discharged by `sorry` is an assertion the kernel accepts on faith, and no statement this paper relies on is discharged that way. Every result that needs an open statement takes it as an explicit hypothesis, so the dependency appears in the type rather than in a comment. The open statements are the identity behind the pair classification, the independence statement $\alpha(G) \geq m$, and the window on $25 \leq m \leq 63$.

Consequently `#print axioms` on any declaration cited in the table above returns `propext`, `Classical.choice` and `Quot.sound`, or a subset of them, and never `sorryAx`. That is a mechanical check, and it is what separates what this paper establishes from what it assumes. No declaration in the Lean modules on which the table above depends carries a `sorry` body. The four stated-first contracts that previously did, for eight runners and for the residual case of four, are held outside it; they were imported by nothing cited here, and they remain conjectural.

9. Where the window statement now stands

The prime case is settled outright, and we state it because the reduction below rests on it.

Theorem 9.1 (Verified)**.** *Let $2m+1$ be prime. Then every set of m distinct positive velocities bounded by $2m$ satisfies* (1)*.*

Proof. The complementary pairs $\{a, 2m+1-a\}$ for $1 \le a \le m$ partition $[1, 2m]$, and each satisfies $\gcd(a, 2m+1) = 1$ because $2m+1$ is prime, so each is a bad pair by Proposition 5.11. A complement W of size m containing no bad set is independent in the bad-pair graph, so it meets each of those m pairs at most once, hence exactly once. It avoids 1 and $m+1$, both bad singletons, so from the pair $\{1, 2m\}$ it takes $2m$ and from the pair $\{m, m+1\}$ it takes m. But $\{m, 2m\}$ is a bad pair, by Proposition 5.10 at $d = m$. So no such W exists, and by Theorem 3.2 the window statement holds. □

The classification of Section 5 reduces the window statement to a single unproved combinatorial claim. We record the reduction because it isolates what remains.

Suppose $V \subseteq [1, 2m]$ has m elements and defeats (1). By Theorem 3.2 its complement $W = [1, 2m] \setminus V$, also of size m, contains no bad set. Three consequences are immediate and each is verified.

(1) W contains neither 1 nor $m+1$, since both are bad singletons.
(2) W contains no bad pair, so W is independent in the bad-pair graph.
(3) If $2m+1$ is prime the bad-pair graph has a perfect matching on $[1, 2m] \setminus \{1, m+1\}$, and the independence number is below m; this is Theorem 9.1, and no such W exists.

So only composite $2m+1$ remains, and there the independence number is at least m: pairs alone cannot finish, and the bad triples must. What is missing is exactly this.

Conjecture 9.2 (Conjecture)**.** Let $2m+1$ be composite. Every m-element subset of $[1, 2m]$ that avoids 1 and $m+1$ and is independent in the bad-pair graph contains a triple of family SUM or of family SUM′.

Conjecture 9.2 implies the window statement for every m, since the prime case is Theorem 9.1 and the two families are verified in Proposition 5.14. It has been checked for $4 \le m \le 30$ by a search that looks for a counterexample directly rather than enumerating candidates, every run terminating without a cap.

Part of the conjecture is governed by classical extremal theory, and we record that because it localises what is genuinely new. Nothing below is proved for a general m: the extremal input is asymptotic with an unspecified threshold, so every statement in this passage is computed over the range stated. A set is *sum-free* if it contains no solution of $a + b = c$ with a, b, c in the set, $a = b$ permitted. The largest sum-free subsets of $[1, 2m]$ have exactly m elements, so a candidate W has precisely the extremal size, and the extremal sets are classified.

Proposition 9.3 (Computed for $5 \le m \le 12$; classical in substance)**.** *For $5 \le m \le 12$ the sum-free subsets of $[1, 2m]$ of size m are exactly three: the odd numbers, the interval $[m, 2m-1]$, and the interval $[m+1, 2m]$. Each of them contains 1 or $m+1$.*

Corollary 9.4 (Computed for $5 \le m \le 20$)**.** *For $5 \le m \le 20$, every m-element $W \subseteq [1, 2m]$ avoiding 1 and $m+1$ contains a solution of $a+b=c$ with $a, b, c \in W$, where $a = b$ is permitted.*

Proof. The odd numbers contain 1, and both intervals contain $m+1$, so no set of the extremal size avoids both; hence W is not sum-free. The range $5 \le m \le 20$ has been verified directly, by searching for a sum-free m-subset of $[1, 2m]$ avoiding 1 and $m+1$ and finding none. At

$m = 4$ the statement fails: $\{2, 3, 7, 8\}$ is sum-free, has four elements in $[1, 8]$, and avoids 1 and 5. □

The case $a = b$ is not excluded. One might expect independence to rule it out, since $a = b$ puts $\{a, 2a\}$ inside W, but a doubling pair need not be a bad pair: MUL requires $2m/3 < a \le m$ and DBL requires $2a < m + 3$, so the band $(m + 3)/2 < a \le 2m/3$ is covered by neither, and there $\{a, 2a\}$ is an edge of no graph. The smallest instances are $a = 9, 10$ at $m = 15$, $a = 10, 11$ at $m = 17$ and $a = 12, 13$ at $m = 20$. So Corollary 9.4 yields a Schur solution but not necessarily three distinct elements.

What Conjecture 9.2 asks beyond Corollary 9.4 is that some sum triple have distinct entries and meet the threshold $2c \ge 3m + 1$ and the coprimality conditions of the two families. Existence of a solution is settled; distinctness and the side conditions are not. That is where the remaining content sits.

We do not claim the extremal input as new. The maximum size $\lceil n/2 \rceil$ of a sum-free subset of $[1, n]$ is classical, and the structure of sum-free subsets of an interval near that size has been studied at length; Tran [2] characterises the sum-free subsets of $[1, n]$ of density at least $2/5 - c$ for an absolute $c > 0$, and our candidates have density exactly $1/2$. The alternative in that characterisation that a large sum-free set is either entirely odd or has $\min A \ge |A|$ gives Proposition 9.3 for all large m, the two intervals being the two sum-free m-subsets of $[m, 2m]$. We verify the range we use rather than track the exact threshold at which the structure theorem takes hold.

Four measurements delimit what a proof must do.

First, family DBL_3 is not needed: no independent set is met by DBL_3 alone at any m checked, while SUM′ alone meets 1, 2 and 9 of them at $m = 10, 16, 22$. Both sum families are required and the third is not.

Second, writing $c = a + b$, family SUM requires $a > 2m - c$ and SUM′ requires $a \le c - m - 1$, so between them they cover every splitting of c exactly when $2c \ge 3m + 1$, and below that threshold the splitting $a = c - m$ is reached by neither. The threshold is therefore sharp.

Third, every candidate does contain sum triples above that threshold, between 3 and 9 of them, with none lacking one for $4 \le m \le 30$. This is not a statement about sum-free sets alone: with independence dropped, the block $\{2, 3, \ldots, m, m + 2\}$ has all its sums too small and is a counterexample at every m. That block carries every pair $\{a, 2a\}$ with $2a \le m$, which is what independence forbids.

Fourth, the coprimality conditions are severe. Of a candidate's sum triples above the threshold, as few as 2 of 36 satisfy the coprimality that SUM or SUM′ also demands, at $m = 22$. The count of usable triples never falls below 2 for $m \ge 13$, but the proportion does not stay bounded away from zero, so a counting argument over the threshold alone will not establish Conjecture 9.2.

10. Open problems

(1) The uniform window statement: does (1) hold for every m and every velocity set inside $[1, 2m]$? Equivalently, by Theorem 3.2, is $\tau(2m) = m + 1$ for every m?

(2) Classify the bad sets of size four and above. Sizes one to three are treated in Section 5; the pair classification is now complete (Theorem 5.7 and Corollary 5.8), and the triples are handled by families that Proposition 5.2 shows are not exhaustive. Nothing is known about size four. The empirical observation of item 4 below suggests they may not matter for the decision procedure, which would be worth proving.

(3) Classify the minimal bad triples that Proposition 5.2 shows lie outside SUM, SUM′ and DBL_3. The two examples given fail for different reasons, one on a boundary of the hypotheses and one on a coprimality condition, which suggests the families need widening rather than supplementing.
(4) Is the empirical observation that irredundant cores use no bad set of size four or more true for every m? If so, the window statement depends only on the sets classified in Section 5.
(5) Give an arithmetic description of the Gallai–Edmonds set D of the bad-pair graph. All candidates known to us fail, as recorded in Section 7.

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